

LLM Benchmark

Nome del Relatore

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Goal and Keypoints

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Goal: create a reliable, scalable and user-friendly benchmark to test the planning capabilities of LLMs on planning problems.

Keypoints

- Problems Classification
- Hardware Platform and LLM Models Selected
- Outputs and Evaluation Metrics

General Structure

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This slide is used to show the structure of the repo, under the root folder we have:

- **analysis:** contains the material needed to write reports and the report itself.
- **archive:** contains material provided by the professor and a PDDL guide written to help understand key features of the PDDL.
- **Benchmark Framework:** main folder containing the work done to create this benchmark.

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analysis/

domain complexity/

notebooks/

Reports/

archive/

Papers/

Planning Domains/

References/

Quick PDDL Guide/

Benchmark Framework/

...

README.md

General Structure II

Now the Benchmark Framework folder is highlighted, note that to simplify not all the sub folders are shown.

- *evaluators*: code for running evaluation and parsing of outputs
- *Leonardo script*: files to gain access to leonardo booster
- *outputs*: store outputs of benchmark
- *prompts*: prompts written to be feed to LLMs
- *runner*: contains code to run some tests with LLMs on the booster
- *scripts*: contains code necessary for the scoring of the inputs
- *tasks*: problems selected
- *utils*: executable of validator
- *results*: heatmaps showing output metric results
- *run benchmark*: file that runs whole working sequence

Benchmark Framework/

```
evaluators/  
Leonardo script/  
outputs/  
prompts/  
runner/  
scripts/  
tasks/  
utils/  
results/  
run benchmark.py
```

Problems' Classification

Classification Purpose

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Purpose: In order to obtain reliable outputs to be evaluated after we for sure need a way to classify the PDDL problems to be fed to the LLMs chosen, since we had available 20 different problems a easy but effective way to classify them was needed to choose wisely the problems to be used.

The main idea behind this Input Classification develops around two main concepts:

- need of a numerical score representy the difficulty of the problem.
- classify problems based on the type of reasoning to be employed to solve the problem.

Implementation: Raw and Numerical Score

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As stated before the idea of using a raw score to classify the difficulty of the problems came up because a relatively simple and repeatable way to evaluate problems was needed; in order to obtain the score the code in *LLM benchmark/scripts/score_domains_complexity.py* has been designed. The main idea behind the computation of the score is to evaluate for each instance of a PDDL problem the presence or the absence of features that we believed could increment or decrement the difficulty of a problem, specifically:

- N. of Objects
- N of possible Actions
- N. of Goals
- N. of Numeric goal conditions
- N. of Numeric functions
- N. of Numeric preconditions
- N. of Numeric effects
- Has metric
- Topology

A raw score for each instance of a problem is therefore computed and then a normalization between instances of the same domain is employed; then for each domain a summary is presented.

Implementation: Numerical Score II

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The raw score for each instance is computed using the following formula:

$$\text{RawScore} = 0.45 \cdot \log(1 + \text{Nactions}) + 0.20 \cdot \log(1 + \text{Ngoals}) + 0.20 \cdot \log(1 + \text{num-score}) + 0.15 \cdot \text{topology}$$

where the num-score is computed by:

$$\text{NumScore} = 1.0 \cdot (\text{N of num effects}) + 2.0 \cdot (\text{N of num preconditions}) + 3.0 \cdot (\text{N of num goal conditions}) + 1.0 \cdot (\text{has metric})$$

A simple example of a domain summary with the scores computed for each instance is shown here on the side, the domain chosen is *counters*. Also the overall domain gets assigned

```
{  
  "domain": "counters",  
  "instances": 20,  
  "total_raw_score": 59.765337,  
  "min_raw_score": 1.820528,  
  "median_raw_score": 3.177554,  
  "mean_raw_score": 2.988267,  
  "max_raw_score": 3.679334,  
  "domain_difficulty_0_100": 18.130829  
},
```

Da qui in poi, il testo riprende a scorrere su tutta la larghezza della slide. Non essendo più dentro una colonna, occuperà lo spazio anche sotto dove prima c'era l'immagine, permettendoti di sfruttare ogni centimetro della diapositiva.

Implementation: Reasoning Type

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Hardware Platform and Models Selected II

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Output and Evaluation Metrics

Output Structure

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Evaluation Metrics

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